Md. Azim Sikder

Electrical and Electronic Engineering Graduate
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Research Interests

Digital VLSI and RTL Design, Digital Logic Architecture, Design Verification, FPGA-Based Hardware Design, and Hardware Security, including quantum-secure authentication.

Education

Bangladesh University of Engineering and Technology (BUET)
B.Sc. in Electrical and Electronic Engineering
CGPA: 3.89/4.00 *June 2026*

VLSI and Digital Design Projects

- **Adaptive SPI Communication Interface (Full VLSI Flow)** *Cadence EDA, SystemVerilog*
  Team project. Designed the SPI slave and 256×32 memory in RTL and built a layered constrained-random SystemVerilog testbench (generator, driver, monitor, scoreboard, coverage) reaching 100% functional coverage; contributed to synthesis and timing constraints. Design met 100 MHz from RTL to GDS (Genus/Innovus, GPDK045; OpenROAD, IHP SG13G2).
- **Microcontroller-Free Smart Toll Collection System** *Digital Logic Design*
  Designed a complete toll collection system implemented entirely in digital logic, without a microcontroller. Submitted to ICECE 2026 [1].
- **Class-A Common-Emitter RF Power Amplifier (433 MHz ISM Band)** *Cadence Virtuoso*
  Designed voltage-feedback biasing and L-C input/output matching networks in gpdk090; achieved 14.14 dB gain, 20.21 dBm saturated output power, and 5.375% peak PAE at 433 MHz.

Other Projects:

- **Machine Learning:** Hand-gesture recognition using a MEMS accelerometer (MPU6050), with time-domain feature extraction and k-nearest-neighbor classification.
- **Embedded Systems:** Smart alert system for natural hazards with real-time P-wave detection for seismic early warning; MopBot autonomous cleaning robot.
- **Power Electronics:** Arduino-controlled MPPT seven-level inverter for solar power generation [2].

Industry Training

VLSI Industry Attachment Program, Siliconova *November 2025*

- Hands-on training across the chip design flow: RTL design, Verilog HDL, SystemVerilog, verification, DFT, ATPG, physical design, analog layout, DRC/LVS, and IC packaging in the Cadence environment.

Conference Papers (Under Review)

[1] **M. A. Sikder** *et al.*, "A Microcontroller-Free Digital Logic Architecture for Smart Toll Collection Systems," submitted to *2026 14th International Conference on Electrical and Computer Engineering (ICECE)*, Dhaka, Bangladesh.

[2] **M. A. Sikder** *et al.*, "An Arduino-Controlled MPPT Seven-Level Inverter for Solar Power Generation," submitted to *2026 14th International Conference on Electrical and Computer Engineering (ICECE)*, Dhaka, Bangladesh.

Undergraduate Thesis

Quantum Token for Anti-Counterfeit Systems *April 2025 – June 2026*
(Under the supervision of Dr. Sajid Muhaimin Choudhury)

- Designed a hybrid quantum-classical authentication scheme combining BB84 encoding with Learning With Errors (LWE), achieving a **100% valid user** and **0% adversary success rate** at **1,500 qubits**, versus the $\sim 4.2 \times 10^6$ qubits required by pure quantum tokens.
- Benchmarked on the **156-qubit IBM Fez** (Heron r2) processor via Qiskit Runtime, with an adaptive verification algorithm that separates hardware noise from tampering.

Technical Skills

HDL: Verilog HDL, SystemVerilog

Programming: C/C++, Python, MATLAB, ARM Assembly, RISC-V Assembly, Qiskit

EDA and Simulation Tools: Cadence (Virtuoso and digital implementation flow), Quartus, PSpice, Proteus, MATLAB/Simulink

Hardware Platforms: FPGA, STM32, Arduino, ESP32

Teaching and Mentoring Experience

Independent Private Tutor *2022 – Present*

- Taught undergraduate Electrical Engineering, Mathematics, and Physics to students from Stony Brook University (USA) and the University of Auckland (New Zealand).
- Supported students with coursework, assignments, and exam preparation.

Scholarships and Awards

- Dean's List Scholarship, BUET
- University Merit Stipend Scholarship, BUET
- Champion, Integration Bee Competition organized by Tachyon (2022)
- Exceptional Mentorship Award, Brain Stormers Academic and Admission Care (2023–2024)

Leadership and Activities

- Organizer and Problem Setter, Integration Bee, EEE Day 2026, BUET
- Member, BUET Robotics Club

References

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